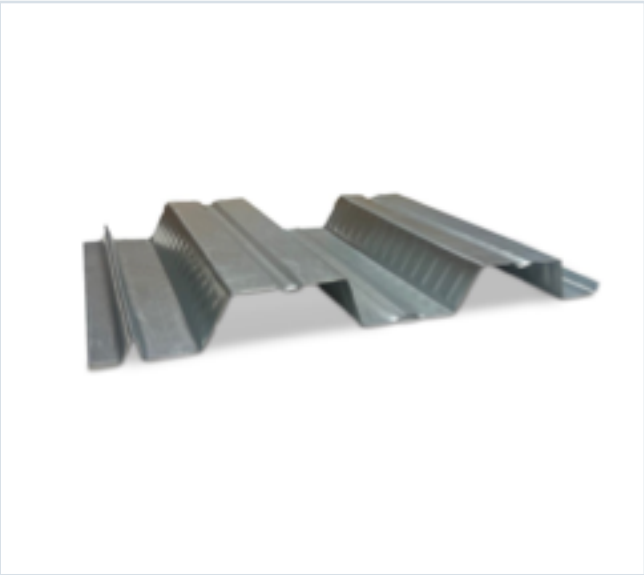


DECKING PRODUCT SUBMITTAL

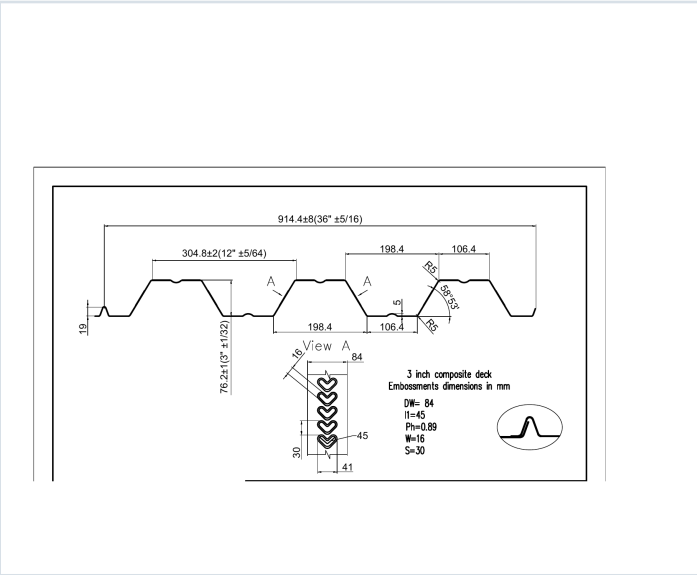
3CD-FY50-16-G90

PRODUCT	GRADE	GAUGE	COATING
3" Composite Deck	Grade 50	16 ga	G90

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 16 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0598	3.5	50	1.009	1.010	30200	30230	1.624	1.610

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 62, 63, 64, 65, 66
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

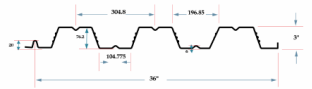
3CD-FY50-16-G90 · 16 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 62

GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (lb)	E _x (ksi)	S _x + S _y (in ³ /ft)	Se (in ³ /ft)	ASD (D ± 16%) M _y /O (in ³ /ft)	M _y /O (in ³ /ft)	I _y + I _x (in ⁴ /ft)	I _y (in ⁴ /ft)	I _x (in ⁴ /ft)
22	0.0295	17	50	0.397	0.426	10709	10747	0.722	0.680	0.970
20	0.0358	21	50	0.525	0.559	13922	13962	1.274	1.260	1.260
18	0.0474	27	50	0.796	0.796	23622	23622	1.274	1.260	1.260
16	0.0598	35	50	1.009	1.005	30000	30000	1.604	1.604	1.604

Note: All section properties and ASD flexural strengths are calculated in accordance with AISI/S100 (2017), AISI S100-2012 and AISI S100-2016.

Shear and Web Crippling

Gage	V _y /O (lb/ft)	Web Crippling (R _y), lb/ft One Flange Loading End Bearing	Web Crippling (R _y), lb/ft One Flange Loading Interior Bearing
22	789	403	375
20	2385	578	754
18	4272	874	1224
16	7370	1497	1864

Note: All section properties and ASD flexural strengths are calculated in accordance with AISI/S100 (2017), AISI S100-2012 and AISI S100-2016.

Allowable Uniform Downward Loads, ASD (PSF)

Span	Gage	6"	7"	8"	9"	10"	12"	14"	16"	18"
Single	22	207	152	124	98	79	68	55	47	38
	20	291	214	164	129	105	87	73	62	53
	18	441	324	248	196	159	131	110	94	81
	16	599	437	335	249	201	166	140	119	103
Double	22	236	173	133	105	85	70	59	50	43
	20	340	256	194	156	127	104	88	76	66
	18	441	324	248	196	159	131	110	94	81
	16	599	437	335	249	201	166	140	119	103
Triple	22	226	173	133	105	85	70	59	50	43
	20	338	255	194	156	127	104	88	76	66
	18	551	405	303	245	198	164	138	117	101
	16	700	514	394	311	252	208	175	149	129

Notes:
1. All section properties and ASD (D ± 16%) uniform loads are calculated in accordance with AISI/S100 (2017), AISI S100-2012 and AISI S100-2016.
2. Loads shown in tables are uniformly distributed downward loads in psf. Span length assumes center-to-center spacing of supports. Tabulated loads shall not be increased by increasing clear span dimensions.
3. Bending Moment formula used for forward stress limitations are: Simple and Two-Span: $M = \frac{wL^2}{8}$; Three-Span: $M = \frac{wL^2}{16}$.
4. Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

STEEL CATALOG PAGE 63

Span	Gage	6"	7"	8"	9"	10"	12"	14"	16"	18"
Single	22	207	152	124	98	79	68	55	47	38
	20	291	214	164	129	105	87	73	62	53
	18	441	324	248	196	159	131	110	94	81
	16	599	437	335	249	201	166	140	119	103
Double	22	236	173	133	105	85	70	59	50	43
	20	340	256	194	156	127	104	88	76	66
	18	441	324	248	196	159	131	110	94	81
	16	599	437	335	249	201	166	140	119	103
Triple	22	226	173	133	105	85	70	59	50	43
	20	338	255	194	156	127	104	88	76	66
	18	551	405	303	245	198	164	138	117	101
	16	700	514	394	311	252	208	175	149	129

Note: For loads that cause L/250 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/160 Deflection, multiply by 0.67.

Construction Span Table – 20 psf Construction Load

Normal Weight Concrete (150 pcf)						Lightweight Concrete (115 pcf)					
Total Slab Depth	Deck Type	Maximum Unfinished Clear Span			Total Slab Depth	Deck Type	Maximum Unfinished Clear Span				
		1 span	2 span	3 span			1 span	2 span	3 span		
16.5 (H-200) 48 PSF	3x12x22 ga	10' 3"	11' 4"	11' 9"	16.5 (H-200) 37 PSF	3x12x22 ga	9' 2"	12' 3"	12' 7"		
	3x12x20 ga	11' 3"	12' 0"	13' 5"		3x12x20 ga	10' 0"	13' 1"	14' 6"		
	3x12x18 ga	13' 5"	15' 6"	16' 0"		3x12x18 ga	14' 3"	16' 8"	17' 3"		
	3x12x16 ga	14' 3"	17' 4"	18' 3"		3x12x16 ga	15' 0"	18' 10"	19' 5"		
16.5 (H-250) 54 PSF	3x12x22 ga	9' 10"	10' 10"	11' 3"	16.5 (H-250) 44 PSF	3x12x22 ga	10' 6"	11' 8"	12' 1"		
	3x12x20 ga	11' 6"	12' 5"	13' 0"		3x12x20 ga	12' 4"	13' 5"	15' 0"		
	3x12x18 ga	12' 5"	14' 3"	14' 9"		3x12x18 ga	13' 2"	15' 2"	16' 8"		
	3x12x16 ga	14' 0"	16' 9"	17' 4"		3x12x16 ga	15' 0"	18' 0"	19' 5"		
16.5 (H-400) 58 PSF	3x12x22 ga	9' 0"	10' 1"	10' 5"	16.5 (H-400) 47 PSF	3x12x22 ga	10' 0"	11' 1"	11' 6"		
	3x12x20 ga	10' 8"	11' 8"	12' 1"		3x12x20 ga	11' 9"	12' 9"	13' 2"		
	3x12x18 ga	12' 0"	13' 1"	14' 3"		3x12x18 ga	13' 2"	15' 2"	16' 8"		
	3x12x16 ga	13' 1"	15' 6"	16' 0"		3x12x16 ga	14' 3"	17' 1"	17' 8"		
16.5 (H-400) 64 PSF	3x12x22 ga	8' 8"	9' 9"	10' 1"	16.5 (H-400) 49 PSF	3x12x22 ga	9' 10"	10' 10"	11' 3"		
	3x12x20 ga	10' 3"	11' 2"	11' 6"		3x12x20 ga	11' 6"	12' 5"	12' 10"		
	3x12x18 ga	11' 6"	13' 3"	13' 9"		3x12x18 ga	12' 0"	14' 10"	15' 4"		
	3x12x16 ga	12' 8"	15' 0"	15' 6"		3x12x16 ga	13' 0"	16' 9"	17' 4"		
16.5 (H-400) 70 PSF	3x12x22 ga	8' 0"	9' 0"	9' 9"	16.5 (H-400) 52 PSF	3x12x22 ga	9' 4"	10' 4"	10' 9"		
	3x12x20 ga	9' 11"	10' 9"	11' 2"		3x12x20 ga	11' 0"	11' 11"	12' 3"		
	3x12x18 ga	11' 4"	12' 10"	13' 4"		3x12x18 ga	12' 4"	14' 2"	14' 8"		
	3x12x16 ga	12' 4"	14' 6"	15' 0"		3x12x16 ga	13' 5"	16' 0"	16' 6"		
16.5 (H-400) 99 PSF	3x12x22 ga	8' 0"	9' 0"	9' 9"	16.5 (H-400) 59 PSF	3x12x22 ga	9' 4"	10' 4"	10' 9"		
	3x12x20 ga	9' 11"	10' 9"	11' 2"		3x12x20 ga	11' 0"	11' 11"	12' 3"		
	3x12x18 ga	11' 4"	12' 10"	13' 4"		3x12x18 ga	12' 4"	14' 2"	14' 8"		
	3x12x16 ga	12' 4"	14' 6"	15' 0"		3x12x16 ga	13' 5"	16' 0"	16' 6"		

Note: Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

PUBLISHED PERFORMANCE DATA

3CD-FY50-16-G90 · 16 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 64

22 ga Normalweight Concrete (145 pcf, f'c = 3,000 psi)

Slab Thickness (Inches)	Weight (psf)	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"
5	46	272	258	242	196	165	147	132
5.5	52	329	288	254	225	200	179	160
6	58	388	340	300	266	237	212	190
6.5	64	400	354	317	308	274	248	223
7	70	400	400	386	351	313	280	252
7.5	76	400	400	400	395	352	316	284

Slab Thickness (Inches)	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"	14'-6"	15'-0"
5	118	127	16	87	79	11	65	59
5.5	144	150	117	106	96	88	80	72
6	171	154	140	127	115	105	95	87
6.5	198	175	162	147	134	122	111	102
7	227	205	186	169	154	140	128	117
7.5	256	235	210	191	174	158	145	132

20 ga Normalweight Concrete (145 pcf, f'c = 3,000 psi)

Slab Thickness (Inches)	Weight (psf)	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"
5	46	329	288	254	225	200	180	162
5.5	52	397	349	308	273	244	218	196
6	58	400	400	363	323	288	258	232
6.5	64	400	400	400	374	334	299	270
7	70	400	400	400	400	381	342	308
7.5	76	400	400	400	400	400	385	347

Slab Thickness (Inches)	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"	14'-6"	15'-0"
5	146	132	115	108	96	80	82	75
5.5	177	160	145	132	120	100	100	92
6	210	180	172	157	143	121	120	110
6.5	244	221	201	185	167	140	140	130
7	278	252	229	209	189	175	160	147
7.5	314	285	259	236	216	198	181	166

18 ga Normalweight Concrete (145 pcf, f'c = 3,000 psi)

Slab Thickness (Inches)	Weight (psf)	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"
5	46	400	400	381	359	303	273	246
5.5	52	400	400	400	397	355	320	289
6	58	400	400	400	400	400	370	334
6.5	64	400	400	400	400	400	400	368
7	70	400	400	400	400	400	400	400
7.5	76	400	400	400	400	400	400	400

Slab Thickness (Inches)	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"	14'-6"	15'-0"
5	223	203	185	169	154	142	130	120
5.5	262	238	217	198	182	162	148	136
6	303	275	251	230	210	193	178	164
6.5	343	314	287	262	240	221	204	188
7	384	354	324	296	272	250	230	213
7.5	400	397	362	331	304	280	258	238

STEEL CATALOG PAGE 66

Slab Thickness (Inches)	Weight (psf)	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"
5	37	400	400	358	338	294	265	240
5.5	42	400	400	400	385	345	313	281
6	47	400	400	400	400	399	369	335
6.5	49	400	400	400	400	400	400	374
7	52	400	400	400	400	400	400	400
7.5	58	400	400	400	400	400	400	400

Slab Thickness (Inches)	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"	14'-6"	15'-0"
5	217	198	181	166	152	140	129	119
5.5	255	232	212	194	178	164	151	140
6	295	269	245	225	207	190	176	162
6.5	339	309	282	259	238	219	203	188
7	384	350	320	294	270	249	230	213
7.5	400	390	357	327	301	278	257	237

16 ga Lightweight Concrete (115 pcf, f'c = 3,000 psi)

Slab Thickness (Inches)	Weight (psf)	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"
5	37	400	400	391	349	313	282	255
5.5	42	400	400	400	400	390	343	310
6	47	400	400	400	400	400	400	369
6.5	49	400	400	400	400	400	400	400
7	52	400	400	400	400	400	400	400
7.5	58	400	400	400	400	400	400	400

Slab Thickness (Inches)	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"	14'-6"	15'-0"
5	231	211	193	177	162	149	138	127
5.5	262	237	215	195	178	162	148	136
6	303	275	250	227	206	187	171	156
6.5	343	314	287	262	239	219	203	188
7	400	400	378	347	320	298	273	253
7.5	400	400	400	391	361	333	308	286